

Reciprocal-Power Sign Rigidity and Stieltjes Moment Lattices

A nonredundant continuation toward the Alaoglu–Erdős conjecture

Research progress report

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Abstract

Let $x \in \mathbb{R}$ satisfy $2^x, 3^x \in \mathbb{Z}$. The conjecture asserts that $x \in \mathbb{Z}$. This report continues the attached unified manuscript without reproducing its surveys of standard determinant, p -adic, interpolation, automaticity, or point-counting approaches. The new work starts from the reciprocal exponent $\theta = 1/x$ and the integer semigroup correspondence

$$A^m B^n \mapsto 2^m 3^n, \quad A = 2^x, \quad B = 3^x.$$

Four concrete advances are obtained. First, every counterexample gives an infinite hierarchy of strict signs of explicitly integral generalized Vandermonde determinants; a wrong sign is therefore a finite, logarithm-free contradiction certificate. Second, the consecutive pure-axis family is factored in closed form. Its sign hierarchy is automatic and therefore does not test compatibility: genuinely mixed minors are indispensable. Third, scalar Stieltjes divided differences are lifted to positive Hankel Gram matrices. After multiplication by one Vandermonde, these matrices are positive definite over $\mathbb{Z}$. Normalizing the underlying Stieltjes measure produces an *output-sensitive integer lattice*: every nonzero integer polynomial has squared L^2 -mass at least the reciprocal of a concrete interpolation determinant. Fourth, the same matrices are connected to the Laurent expansion of the rational interpolation function and hence to Padé and subresultant calculations, giving a fraction-free route for exact search and formal verification.

The report also derives scale reduction, mixed concavity inequalities, a tail-localization lemma, integer-Chebyshev contradiction criteria, an exact finite search protocol, and a Lean-oriented certificate format. These results do not yet close the conjecture. The remaining bottleneck is stated sharply: one must construct mixed semigroup node sets for which analytic concentration beats the output-dependent lattice threshold, or exhibit a uniformly wrong mixed subresultant sign for every non-power pair (A, B) .

Status discipline. Every assertion below is labelled by type. “Proved” means a complete argument is given. “Exact experiment” means integer arithmetic on a finite set, not floating-point evidence. “Criterion” is a rigorously sufficient condition whose universal verification remains open. No unconditional proof of the Alaoglu–Erdős conjecture is asserted.

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1 What is new in this continuation

The attached report already assembles a broad collection of routes toward the conjecture. Repeating those reductions would add bulk without adding leverage. The present continuation therefore uses only the minimal normal form needed to establish a different interface: positive integral semigroup data are regarded as samples of the Bernstein function $u \mapsto u^\theta$, with $0 < \theta < 1$. The scalar sign information is then promoted to quadratic forms, lattices, and subresultants.

The progress ledger is as follows.

Component	Content	Status
Counterexample normal form	$x > 1$ transcendental; $A = 2^x$, $B = 3^x$, $\theta = 1/x \in (0, 1)$	proved
Integer sign oracle	Strict signs of mixed generalized Vandermonde determinants	proved
Axis factorization	Closed forms on $\{A^j\}$ and $\{B^j\}$; proof that axis-only tests do not couple	proved
Moment lattice	Positive definite Hankel matrix becomes integral after one Vandermonde factor	proved
Output normalization	Every nonzero $P \in \mathbb{Z}[s]$ has $\ P\ _{L^2(\mu)}^2 \geq 1/k_0$	proved
Padé/subresultant bridge	Fraction-free algebraic representation of all moment minors	proved as an algorithmic bridge
Tail and Chebyshev tests	Explicit sufficient inequalities for contradiction	conditional criteria
Finite search	Exact determinant and moment-minor filters over a bounded candidate box	exact experiment
Global conclusion	Uniform mixed certificate for every non-power pair	open

A useful conceptual change is that the arithmetic lower bound is no longer merely “a nonzero rational has size at least the reciprocal of its denominator.” The normalized Stieltjes construction gives a whole positive quadratic form on the coefficient lattice $\mathbb{Z}^{d+1}$. The denominator is the integer k_0 determined by the *outputs*, and every nonzero lattice vector has norm at least $1/k_0$. This is the main new object.

2 Minimal counterexample normal form

Proposition 2.1 (Elementary reduction). *Assume $2^x, 3^x \in \mathbb{Z}$. Then either $x \in \{0, 1\}$ or $x > 1$. If $x \in \mathbb{Q}$, then $x \in \mathbb{Z}$. Consequently, a nonintegral counterexample must have $x > 1$ and must be transcendental.*

Proof. If $x < 0$, then $0 < 2^x < 1$, so 2^x is not an integer. If $0 < x < 1$, then $1 < 2^x < 2$, again impossible. Now write a positive rational $x = p/q$ in lowest terms. If $A = 2^{p/q} \in \mathbb{Z}$, then $A^q = 2^p$. Taking the 2-adic valuation gives $qv_2(A) = p$, hence $q \mid p$ and $q = 1$. Finally, if a

nonintegral x were algebraic, the Gelfond–Schneider theorem would make 2^x transcendental, contrary to the hypothesis. $\square$

For the rest of the report, assume hypothetically that a nonintegral counterexample exists and put

$$A = 2^x \in \mathbb{N}, \quad B = 3^x \in \mathbb{N}, \quad \theta = \frac{1}{x} \in (0, 1). \quad (1)$$

Then

$$A^\theta = 2, \quad B^\theta = 3. \quad (2)$$

The passage to θ is not cosmetic: all derivatives of u^θ have a fixed alternating sign after the first derivative, independently of the size of x .

Definition 2.2 (Semigroup dictionary). For $e = (m, n) \in \mathbb{N}_0^2$, define

$$U_e = A^m B^n, \quad V_e = 2^m 3^n.$$

Then $U_e = V_e^x$ and $V_e = U_e^\theta$.

Unique factorization makes the V_e distinct. The same is true of the U_e : an equality $A^m B^n = A^{m'} B^{n'}$ would, after taking the positive θ -th power, give $2^m 3^n = 2^{m'} 3^{n'}$. Moreover,

$$U_e < U_{e'} \iff V_e < V_{e'}, \quad (3)$$

because $u \mapsto u^\theta$ is strictly increasing. Thus the order of all input nodes is known from the corresponding 3-smooth outputs, without evaluating a logarithm.

3 A logarithm-free integer sign oracle

Let $f(u) = u^\theta$. For distinct positive numbers $u_0 < \dots < u_k$, write $[u_0, \dots, u_k]f$ for the ordinary divided difference.

Theorem 3.1 (Integral mixed-minor sign hierarchy). *Let $k \geq 1$. Choose distinct exponent vectors $e_0, \dots, e_k \in \mathbb{N}_0^2$ and order them so that $V_{e_0} < \dots < V_{e_k}$. Set $u_i = U_{e_i}$ and $v_i = V_{e_i}$. Under the counterexample hypothesis,*

$$N(e_0, \dots, e_k) := \det \begin{pmatrix} 1 & u_0 & u_0^2 & \dots & u_0^{k-1} & v_0 \\ 1 & u_1 & u_1^2 & \dots & u_1^{k-1} & v_1 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 1 & u_k & u_k^2 & \dots & u_k^{k-1} & v_k \end{pmatrix} \quad (4)$$

is a nonzero integer and

$$\operatorname{sgn} N(e_0, \dots, e_k) = (-1)^{k-1}. \quad (5)$$

In particular, one wrong sign or one zero determinant is a finite exact certificate that (A, B) cannot arise from a counterexample.

Proof. The generalized Vandermonde identity gives

$$N(e_0, \dots, e_k) = \left(\prod_{0 \leq i < j \leq k} (u_j - u_i) \right) [u_0, \dots, u_k]f.$$

The product is positive by (3). The generalized mean-value theorem for divided differences gives a point $\xi \in (u_0, u_k)$ for which

$$[u_0, \dots, u_k]f = \frac{f^{(k)}(\xi)}{k!}.$$

For $0 < \theta < 1$,

$$f^{(k)}(u) = \theta(\theta - 1) \cdots (\theta - k + 1)u^{\theta-k},$$

whose sign is $(-1)^{k-1}$. Strictness follows because none of the factors vanishes. Every entry in (4) is an integer. $\square$

This theorem replaces an equality of logarithmic ratios by an infinite collection of statements in ordered rings. It is especially convenient for proof assistants: after the generic real-analysis lemma is formalized once, each proposed certificate reduces to normalization of natural-number powers and an integer determinant.

3.1 Scaling removes a large class of redundant searches

Proposition 3.2 (Semigroup scaling law). *Let $c = U_e$ and $c^\theta = V_e$. For any $k + 1$ nodes $u_0, \dots, u_k$ from the semigroup,*

$$[cu_0, \dots, cu_k]f = \frac{V_e}{c^k} [u_0, \dots, u_k]f.$$

If $N(\mathcal{U})$ denotes the determinant numerator in (4), then

$$N(c\mathcal{U}) = V_e c^{k(k-1)/2} N(\mathcal{U}). \quad (6)$$

Hence scaling a certificate by a common semigroup element never changes its sign.

Proof. Homogeneity gives $f(cu) = c^\theta f(u)$, while a divided difference of order k acquires c^{-k} under dilation of all nodes. The Vandermonde acquires $c^{k(k+1)/2}$; multiplying the two scaling factors yields (6). $\square$

Thus a search may subtract the componentwise minimum of all exponent vectors and retain only *primitive* patterns touching at least one coordinate axis. This is a substantial reduction in exact enumeration and prevents inflated evidence from repeated copies of the same certificate.

4 Pure-axis factorization and the necessity of mixed nodes

The next result is a negative theorem in the useful sense: it identifies a whole family of methods that cannot couple the two hypotheses.

Theorem 4.1 (Closed factorization on one generator). *For every integer $k \geq 1$,*

$$[1, A, A^2, \dots, A^k]f = \frac{\prod_{r=0}^{k-1} (2 - A^r)}{\prod_{r=0}^{k-1} (A^k - A^r)}, \quad (7)$$

$$[1, B, B^2, \dots, B^k]f = \frac{\prod_{r=0}^{k-1} (3 - B^r)}{\prod_{r=0}^{k-1} (B^k - B^r)}. \quad (8)$$

Both expressions automatically have sign $(-1)^{k-1}$ for every $A > 2$ and every $B > 3$. Consequently this consecutive pure-axis sign hierarchy cannot test the compatibility of $A^\theta = 2$ with $B^\theta = 3$.

Proof. For the A -axis, the determinant numerator has rows indexed by $i = 0, \dots, k$ and columns $1, A^i, A^{2i}, \dots, A^{(k-1)i}, 2^i$. It is therefore the ordinary Vandermonde determinant in the column bases

$$1, A, A^2, \dots, A^{k-1}, 2.$$

Hence it equals

$$\left(\prod_{0 \leq r < s \leq k-1} (A^s - A^r) \right) \left(\prod_{r=0}^{k-1} (2 - A^r) \right).$$

The node Vandermonde contains the same first product and an additional factor $\prod_{r=0}^{k-1} (A^k - A^r)$, proving (7). The B -axis is identical with 2 replaced by 3. The sign count is immediate. $\square$

Formula (7) may also be written as a basic-hypergeometric Newton coefficient. With $q = A^{-1}$, the k -th Newton term at an integer Z is

$$A^{-k} \frac{(2; q)_k (Z; q)_k}{(q; q)_k}, \quad (a; q)_k = \prod_{r=0}^{k-1} (1 - aq^r). \quad (9)$$

This exact cancellation explains why a naive denominator estimate on a pure geometric axis is misleading: most of the generalized Vandermonde has already cancelled. It also gives a calibration family against which every proposed mixed estimate should be tested.

5 The first mixed constraints

The first nontrivial information appears when the two axes are interlaced. Since $1 < 2 < 3 < 4$, the corresponding input nodes satisfy

$$1 < A < B < A^2.$$

Strict concavity of f at the first three and the last three nodes gives the following.

Proposition 5.1 (Initial rational cone). *Every counterexample pair (A, B) must satisfy*

$$B \geq 2A, \quad 2B \leq A^2 + A - 1. \quad (10)$$

Proof. The consecutive secant slopes at $(1, 1), (A, 2), (B, 3)$ strictly decrease, so $1/(A-1) > 1/(B-A)$ and $B > 2A - 1$. Integrality gives $B \geq 2A$. Applying the same argument to $(A, 2), (B, 3), (A^2, 4)$ gives $1/(B-A) > 1/(A^2-B)$, hence $2B < A^2 + A$, and integrality gives the second inequality. $\square$

The order-three sign already adds a nonlinear condition.

Proposition 5.2 (First mixed cubic numerator). *Define*

$$F_3(A, B) = -2A(A-1)^2(A+1) + A(A-1)(A+1)(B-1) - (A-2)(B-1)(B-A). \quad (11)$$

Then every counterexample pair satisfies $F_3(A, B) > 0$.

Proof. Let P_2 be the quadratic interpolant to f at $1, A, A^2$. Newton interpolation and (7) give

$$P_2(z) = 1 + \frac{z-1}{A-1} + \frac{(2-A)(z-1)(z-A)}{A(A-1)^2(A+1)}.$$

The interpolation error at $B \in (A, A^2)$ is

$$3 - P_2(B) = [1, A, B, A^2]f (B-1)(B-A)(B-A^2).$$

The divided difference is positive, while the product is negative. Thus $P_2(B) > 3$. Clearing the positive denominator $A(A-1)^2(A+1)$ gives (11). $\square$

Higher canonical constraints are obtained from the ordered 3-smooth sequence

$$1, 2, 3, 4, 6, 8, 9, 12, 16, 18, 24, 27, 32, 36, \dots$$

by replacing each $2^m 3^n$ with $A^m B^n$ and imposing (5) on every selected subset. The result is a nested sequence of semialgebraic regions in the integer lattice. Unlike a numerical comparison of logarithms, every exclusion comes with a small list of exponent pairs and an integer determinant that can be checked independently.

6 From scalar Stieltjes differences to positive Gram matrices

The function $f(u) = u^\theta$, $0 < \theta < 1$, is a complete Bernstein function. Its elementary Stieltjes representation is

$$u^\theta = c_\theta \int_0^\infty \frac{u}{u+s} s^{\theta-1} ds, \quad c_\theta = \frac{\sin(\pi\theta)}{\pi}. \quad (12)$$

For $n \geq 1$ and distinct positive nodes $u_0, \dots, u_n$, taking an order- n divided difference under the integral gives

$$(-1)^{n-1} [u_0, \dots, u_n]f = c_\theta \int_0^\infty \frac{s^\theta}{\prod_{i=0}^n (s+u_i)} ds > 0. \quad (13)$$

The attached report uses scalar divided-difference positivity. The new step here is to insert polynomial squares into the same measure and keep exact control of their arithmetic.

Fix an ordered semigroup node set

$$\mathcal{U} = (u_0 < \dots < u_n), \quad v_i = u_i^\theta \in \mathbb{N},$$

and an integer $d \geq 0$ with

$$n \geq 2d + 1. \quad (14)$$

For $0 \leq r \leq 2d$, define

$$h_r(\mathcal{U}) = c_\theta \int_0^\infty \frac{s^{\theta+r}}{\prod_{i=0}^n (s+u_i)} ds. \quad (15)$$

Condition (14) guarantees convergence at infinity.

Lemma 6.1 (Rational moment formula). *Let $P_{\mathcal{U}}(z) = \prod_{i=0}^n (z - u_i)$. Then*

$$h_r(\mathcal{U}) = (-1)^{n+r+1} \sum_{i=0}^n \frac{u_i^r v_i}{P'_{\mathcal{U}}(u_i)} = (-1)^{n+r+1} [u_0, \dots, u_n] z^{\theta+r}. \quad (16)$$

In particular, every $h_r(\mathcal{U})$ is rational and positive.

Proof. For a complex parameter β in the convergence strip $-1 < \Re \beta < n$, consider

$$I(\beta) = \int_0^\infty \frac{s^\beta}{\prod_i (s + u_i)} ds.$$

In the substrip $-1 < \Re \beta < 0$, partial fractions may be integrated term by term, using

$$\int_0^\infty \frac{s^\beta}{s + u} ds = -\frac{\pi u^\beta}{\sin \pi \beta}.$$

Since the coefficient of $(s + u_i)^{-1}$ is $(-1)^n / P'_{\mathcal{U}}(u_i)$, one obtains

$$I(\beta) = (-1)^{n+1} \frac{\pi}{\sin \pi \beta} \sum_i \frac{u_i^\beta}{P'_{\mathcal{U}}(u_i)}.$$

Both sides continue analytically through the full convergence strip, with removable singularities at integral parameters after the sum is formed. Put $\beta = \theta + r$ and use $\sin \pi(\theta + r) = (-1)^r \sin \pi \theta$. $\square$

7 The output-dependent moment lattice

Let

$$\Delta(\mathcal{U}) = \prod_{0 \leq i < j \leq n} (u_j - u_i) > 0, \quad k_r(\mathcal{U}) = \Delta(\mathcal{U}) h_r(\mathcal{U}). \quad (17)$$

Theorem 7.1 (Integral positive moment matrix). *Under the counterexample hypothesis and (14), every $k_r(\mathcal{U})$ is a positive integer. The Hankel matrix*

$$K_d(\mathcal{U}) = (k_{i+j}(\mathcal{U}))_{0 \leq i, j \leq d} \quad (18)$$

is an integral positive-definite matrix. Consequently,

$$\det K_d(\mathcal{U}) \in \mathbb{N}, \quad \det H_d(\mathcal{U}) \geq \Delta(\mathcal{U})^{-(d+1)}, \quad (19)$$

where $H_d = (h_{i+j})$.

Proof. By (16),

$$k_r = (-1)^{n+r+1} \sum_i u_i^r v_i \frac{\Delta(\mathcal{U})}{P'_{\mathcal{U}}(u_i)}.$$

The quotient $\Delta(\mathcal{U})/P'_{\mathcal{U}}(u_i)$ is an integer: after cancellation it is, up to sign, the product of all pairwise differences not incident with u_i . Positivity follows from the integral definition (15).

For a nonzero vector $c = (c_0, \dots, c_d)^T \in \mathbb{R}^{d+1}$ and $Q_c(s) = \sum_{j=0}^d c_j s^j$,

$$c^T K_d c = \Delta(\mathcal{U}) c_\theta \int_0^\infty \frac{s^\theta Q_c(s)^2}{\prod_i (s + u_i)} ds > 0.$$

Thus K_d is positive definite. Its determinant is a positive integer, so it is at least one. Since $K_d = \Delta(\mathcal{U}) H_d$, (19) follows. $\square$

The normalization reveals the promised output sensitivity. Put

$$k_0(\mathcal{U}) = \Delta(\mathcal{U})h_0(\mathcal{U}) = \left| \det \begin{pmatrix} 1 & u_0 & \cdots & u_0^{n-1} & v_0 \\ \vdots & \vdots & & \vdots & \vdots \\ 1 & u_n & \cdots & u_n^{n-1} & v_n \end{pmatrix} \right| \in \mathbb{N} \quad (20)$$

and define the probability measure

$$d\mu_{\mathcal{U}}(s) = \frac{c_{\theta}}{h_0(\mathcal{U})} \frac{s^{\theta}}{\prod_{i=0}^n (s + u_i)} ds. \quad (21)$$

Corollary 7.2 (Output-dependent L^2 lattice). *For every nonzero polynomial $Q \in \mathbb{Z}[s]$ of degree at most d ,*

$$\int_0^\infty Q(s)^2 d\mu_{\mathcal{U}}(s) = \frac{M_{\mathcal{U}}(Q)}{k_0(\mathcal{U})} \geq \frac{1}{k_0(\mathcal{U})}, \quad (22)$$

where $M_{\mathcal{U}}(Q)$ is a positive integer. In addition, if $G_d = (h_{i+j}/h_0)_{0 \leq i,j \leq d}$ is the normalized Gram matrix, then

$$\det G_d(\mathcal{U}) = \frac{\det K_d(\mathcal{U})}{k_0(\mathcal{U})^{d+1}} \geq k_0(\mathcal{U})^{-(d+1)}. \quad (23)$$

Proof. If $c \in \mathbb{Z}^{d+1} \setminus \{0\}$ is the coefficient vector of Q , then $c^T K_d c$ is a positive integer. Divide by $k_0 = \Delta h_0$. The determinant identity is immediate. $\square$

The denominator in (22) is not a generic common denominator. It is the interpolation determinant formed from the actual output values $2^m 3^n$. This is the point at which cancellation specific to the conjecture enters the moment method.

7.1 Unimodular recentering

For an integer C , the bases $1, s, \dots, s^d$ and $1, (s - C), \dots, (s - C)^d$ are related by a unitriangular integral matrix. Therefore

$$\det(\mathbb{E}_{\mu}[(S - C)^{i+j}])_{0 \leq i,j \leq d} = \det G_d(\mathcal{U}). \quad (24)$$

This permits analytic estimates around an integer center without weakening the arithmetic lower bound. More generally, every unimodular change of polynomial basis preserves the lattice determinant.

7.2 Smith and divisibility refinements

The lower bound “at least one” for $\det K_d$ is deliberately conservative. In exact examples, the matrix often has large systematic divisors. A fraction-free Smith-normal-form calculation yields invariant factors

$$s_1 \mid s_2 \mid \cdots \mid s_{d+1}, \quad \det K_d = s_1 \cdots s_{d+1}.$$

Any uniform divisor extracted from the mixed output pattern strengthens (23). This is one of the most concrete current targets: pure-axis divisors factor completely by [theorem 4.1](#), while mixed invariant factors can detect compatibility.

8 Interpolation rational functions and subresultants

Let $R_{\mathcal{U}}(z)$ be the unique polynomial of degree at most n satisfying $R_{\mathcal{U}}(u_i) = v_i$. The partial fraction expansion gives

$$\frac{R_{\mathcal{U}}(z)}{P_{\mathcal{U}}(z)} = \sum_{i=0}^n \frac{v_i}{P'_{\mathcal{U}}(u_i)(z - u_i)} = \sum_{r \geq 0} m_r z^{-r-1}, \quad (25)$$

where

$$m_r = \sum_{i=0}^n \frac{u_i^r v_i}{P'_{\mathcal{U}}(u_i)}, \quad h_r = (-1)^{n+r+1} m_r. \quad (26)$$

Thus the positive moment matrices are signed Hankel blocks of the Laurent expansion at infinity of a rational interpolation function determined entirely by integer input-output data.

Proposition 8.1 (Fraction-free Padé bridge). *After clearing the single factor $\Delta(\mathcal{U})$, the leading consecutive Hankel determinants used in [theorem 7.1](#) can be computed in either of two equivalent exact ways, with the Laurent expansion taken at infinity and the corresponding reversed-polynomial sign conventions fixed:*

- (i) *from the integral-moment formula [\(16\)](#) followed by a Bareiss determinant;*
- (ii) *from the Padé table at infinity of $R_{\mathcal{U}}/P_{\mathcal{U}}$, equivalently from the signed subresultant polynomial-remainder sequence of $P_{\mathcal{U}}$ and $R_{\mathcal{U}}$.*

Nonvanishing of the relevant Hankel determinants is equivalent to normality of the corresponding Padé entries. Under a counterexample, the convention-adjusted signs are fixed by positivity.

Proof sketch. Equation [\(25\)](#) identifies the Laurent coefficients. The denominator of a Padé approximant is obtained by solving a Hankel linear system in these coefficients; its solvability and degree are controlled by consecutive Hankel minors. The classical Hankel-subresultant identity identifies the same systems with principal minors in the Sylvester/subresultant construction for $P_{\mathcal{U}}$ and $R_{\mathcal{U}}$. Because $P_{\mathcal{U}}$ is monic and $\Delta(\mathcal{U})R_{\mathcal{U}} \in \mathbb{Z}[z]$, fraction-free elimination remains in $\mathbb{Z}$ throughout. See [\[13, 4, 9\]](#) for the standard identities. $\square$

This bridge matters for two reasons. First, subresultant signs can be certified without approximating θ . Second, a proof assistant need not formalize all of Padé theory to check a discovered certificate: the final object is simply an integer determinant or a fraction-free remainder with a specified sign.

9 Concrete contradiction criteria

The moment lattice converts the conjecture into a competition between an analytic upper bound and an arithmetic lower bound.

Criterion 9.1 (Single-polynomial certificate). *A counterexample is impossible if one can find a semigroup node set $\mathcal{U}$ with $n \geq 2d + 1$ and a nonzero $Q \in \mathbb{Z}[s]$, $\deg Q \leq d$, such that*

$$c_{\theta} \int_0^{\infty} \frac{s^{\theta} Q(s)^2}{\prod_i (s + u_i)} ds < \frac{1}{\Delta(\mathcal{U})}. \quad (27)$$

Equivalently, it is enough to prove $\int Q^2 d\mu_{\mathcal{U}} < 1/k_0(\mathcal{U})$.

This is immediate from [theorem 7.1](#) and [corollary 7.2](#), but it is useful because the left side can be bounded without expanding the enormous determinant k_0 .

Criterion 9.2 (Determinant certificate). *A counterexample is impossible if, for some $\mathcal{U}$ and d ,*

$$\det(\mathbb{E}_\mu[(S - C)^{i+j}])_{0 \leq i, j \leq d} < k_0(\mathcal{U})^{-(d+1)} \quad (28)$$

for one (hence every) integer center C . By Hadamard's inequality it is enough that

$$\prod_{j=0}^d \mathbb{E}_\mu[(S - C)^{2j}] < k_0(\mathcal{U})^{-(d+1)}. \quad (29)$$

The determinant version can succeed even when no explicit short polynomial is guessed: a geometry-of-numbers or LLL step can then recover one. Conversely, the exact integral matrix K_d provides a direct check that a numerical short vector is not an artefact of rounding.

9.1 A tail-localization lemma

Write

$$W_{\mathcal{U}}(s) = \frac{s^\theta}{\prod_i (s + u_i)}, \quad N_{\mathcal{U}}(s) = \#\{i : u_i \leq s\}.$$

Lemma 9.3 (Node-count tail contraction). *For $\lambda \geq 1$ and $s > 0$,*

$$\frac{W_{\mathcal{U}}(\lambda s)}{W_{\mathcal{U}}(s)} \leq \lambda^\theta \left(\frac{2}{\lambda + 1} \right)^{N_{\mathcal{U}}(s)}. \quad (30)$$

Consequently, for every $R > 0$,

$$\int_{\lambda R}^{\lambda^2 R} W_{\mathcal{U}}(s) \, ds \leq \lambda^{\theta+1} \left(\frac{2}{\lambda + 1} \right)^{N_{\mathcal{U}}(R)} \int_R^{\lambda R} W_{\mathcal{U}}(s) \, ds. \quad (31)$$

Proof. The ratio is

$$\lambda^\theta \prod_i \frac{s + u_i}{\lambda s + u_i}.$$

If $u_i \leq s$, put $t = u_i/s \in (0, 1]$. Then $(1 + t)/(\lambda + t) \leq 2/(\lambda + 1)$; the remaining factors are at most one. For (31), substitute $s = \lambda t$ and use $N_{\mathcal{U}}(t) \geq N_{\mathcal{U}}(R)$ on $[R, \lambda R]$. $\square$

For a node set consisting of an initial segment of the mixed semigroup, $N_{\mathcal{U}}(s)$ grows stepwise with the number of available 3-smooth values. Iterating (31) therefore gives rigorous tail bounds that improve as additional mixed nodes are inserted. The unresolved comparison is whether this concentration can outrun the simultaneous growth of $k_0(\mathcal{U})$.

9.2 An integer-Chebyshev target on a unit interval

For an integer C , the polynomial

$$T_C(s) = (s - C)(s - C - 1)$$

has integer coefficients and $|T_C(s)| \leq 1/4$ on $[C, C+1]$. Hence $Q_m = T_C^m$ obeys $|Q_m|^2 \leq 16^{-m}$ there. Splitting the normalized integral gives

$$\|Q_m\|_{L^2(\mu_U)}^2 \leq 16^{-m} + \int_{[0,\infty) \setminus [C,C+1]} |T_C(s)|^{2m} d\mu_U(s). \quad (32)$$

Combining (32) with lemma 9.3 gives a fully explicit route to criterion 9.1. It is not enough to show mere concentration: the tail must be controlled with the polynomial growth included, and the resulting bound must beat $1/k_0$. Here $\deg Q_m = 2m$, so an application of the moment lattice requires $d \geq 2m$ and $n \geq 2d + 1$; choosing $d = 2m$ requires $n \geq 4m + 1$.

Open problem 9.4 (Quantitative localization problem). Construct, from every hypothetical non-power pair (A, B) , primitive mixed node sets $\mathcal{U}_j$, integer intervals $[C_j, C_j + 1]$, and degrees m_j for which the right side of (32) is $o(1/k_0(\mathcal{U}_j))$.

Integral-exponent controls $(A, B) = (2^r, 3^r)$ show that a bound depending only on crude node sizes cannot solve open problem 9.4; it must use mixed arithmetic, for example Smith invariant factors or unusually small mixed subresultants.

10 Exact finite experiments

All computations in this section use Python integers, the fraction-free Bareiss determinant, and exact exponentiation. No logarithm is used in a certificate. Floating point is used only to select a few illustrative near-power pairs for display; their acceptance or rejection is then decided exactly.

10.1 Bounded mixed-sign hierarchy

The experiment enumerates

$$3 \leq A \leq 60, \quad 2A \leq B \leq \frac{A^2 + A - 1}{2},$$

then imposes order agreement on the first twelve 3-smooth outputs and all subset determinant signs through a fixed order. Common-power controls are pairs $(2^r, 3^r)$ in the box.

The exact enumeration fragment was unavailable at build time.

Machine-generated summary. The computation is an exact finite experiment; no global conclusion is inferred from it.

The experiment is deliberately interpreted conservatively. A surviving non-control pair is not evidence for a solution; it only means that the tested finite sign fragment has not yet located a contradiction. An excluded pair, by contrast, comes with a finite exact witness.

10.2 Sample exact certificates

For each displayed pair, the program first checks semigroup order and then searches all subsets of the first fourteen smooth outputs through order six.

The sample-certificate fragment was unavailable at build time.

The pure-power controls are an important guard against sign-convention errors. They must pass every test because they correspond to the genuine exponent $x \in \mathbb{Z}$.

10.3 Moment-minor filters

Using the first eight smooth outputs 1, 2, 3, 4, 6, 8, 9, 12, one has $n = 7$ and may build the 4×4 matrix K_3 . The following table reports the signs of its leading principal minors.

The moment-matrix fragment was unavailable at build time.

A nonpositive leading minor is a compact contradiction certificate. Positive leading minors through this size do not establish the existence of a power exponent; they merely show that this particular truncated Stieltjes moment test has not failed.

10.4 Reproducible pseudocode

```

for each integer pair (A,B) in the rational cone:
  E := exponent pairs of the first N numbers  $2^m 3^n$ 
  U[e] :=  $A^m B^n$ ; V[e] :=  $2^m 3^n$ 
  reject if U is not ordered as V
  for k = 2, ..., K:
    for every selected (k+1)-subset I of E:
      D := det( 1, U, ..., U(k-1), V ) on I
      reject unless sign(D) =  $(-1)^{(k-1)}$ 

for a moment certificate with n+1 nodes and n >= 2d+1:
  Delta := product_{i<j} (U[j]-U[i])
  k[r] :=  $(-1)^{(n+r+1)} \sum_i U[i]^r V[i] \text{Delta}/P'(U[i])$ 
  K := Hankel(k[0], ..., k[2d])
  reject unless all required principal minors of K are positive

```

The determinant numerator itself, together with the exponent vectors, is a portable certificate. A checker does not need the search code.

11 A third-base bridge and why it remains attractive

The Six Exponentials Theorem gives a clean strategic target. If x is irrational, then $1, x$ are linearly independent over $\mathbb{Q}$, while $\log 2, \log 3, \log 5$ are linearly independent over $\mathbb{Q}$. Therefore at least one of

$$2, 3, 5, 2^x, 3^x, 5^x$$

is transcendental. Under a counterexample, the first five numbers except 5^x are algebraic, so the theorem forces 5^x to be transcendental. Consequently:

Criterion 11.1 (Third-base bootstrap). *Any argument deriving the algebraicity of 5^x from $2^x, 3^x \in \mathbb{Z}$ proves the Alaoglu–Erdős conjecture. The same holds with 5 replaced by any positive algebraic number $c \neq 1$ whose logarithm is $\mathbb{Q}$ -independent of $\log 2$ and $\log 3$.*

The moment framework suggests one possible bridge: interpolate $y \mapsto y^x$ at many 3-smooth integers, where all values are integral, and evaluate at 5. What is missing is an arithmetic approximation strong enough to force algebraicity, not merely a good real approximation. Ordinary Lagrange denominators grow too rapidly. A successful construction would likely need one of the output-specific cancellations isolated in [theorem 4.1](#), [corollary 7.2](#), and [proposition 8.1](#).

12 Routes tested in this continuation and the precise obstruction

This section records three tempting diversions so that subsequent work does not restart them without addressing the stated obstruction.

12.1 Dense rational graph

If a counterexample exists, then for every $m, n \in \mathbb{Z}$,

$$(2^{m+nx}, 3^{m+nx}) = (2^m A^n, 3^m B^n) \in \mathbb{Q}_{>0}^2.$$

Thus the real analytic curve $t \mapsto (2^t, 3^t)$ contains a rank-two subgroup of rational points that is dense along the real one-parameter subgroup. This is an appealing functional-transcendence picture, but a direct Pila–Wilkie count is too weak: points in a fixed compact parameter interval obtained with $|m|, |n| \leq N$ have height exponential in N and number only $O(N)$, i.e. logarithmic in height. This lies far below a generic H^ε upper bound. A useful return to this route would require a sharp polylogarithmic count specific to irrational power curves.

12.2 p -adic automatic continuity

The group homomorphism generated by $2 \mapsto 3$ and $A \mapsto B$ is real-order preserving. It is tempting to extend it continuously to a p -adic closure and compare p -adic logarithm ratios. The missing implication is continuity: a congruence $2^m A^n \equiv 1 \pmod{p^r}$ gives no known congruence for $3^m B^n$. Real closeness and p -adic closeness pull in unrelated directions. A future use of this idea needs an independent height-to-congruence transfer principle.

12.3 Arithmetic progressions of smooth bases

The values of y^x are integral at every 3-smooth y . Primitive arithmetic progressions such as 1, 2, 3, 4 and 2, 9, 16 yield exact convexity inequalities in A and B , but known S -unit finiteness prevents an arbitrarily long supply of such stencils. Their best role is as low-cost filters and seeds for mixed determinants, not as a stand-alone route to a third base.

13 Formalization blueprint

The sign certificates were designed to separate analysis from computation.

Layer 1. Generic analysis. Formalize that for $0 < \theta < 1$, the order- k divided difference of u^θ on increasing positive nodes has sign $(-1)^{k-1}$. A generalized mean-value theorem is sufficient; no Stieltjes integration is needed for the first certificate checker.

Layer 2. Semigroup specialization. From $A^\theta = 2$ and $B^\theta = 3$, prove $(A^m B^n)^\theta = 2^m 3^n$ and transfer output order to input order.

Layer 3. Integer determinant. Identify the divided-difference numerator with the generalized Vandermonde determinant and cast its entries into $\mathbb{Z}$.

Layer 4. Certificate evaluation. For a fixed exponent list, use kernel reflection or a fraction-free determinant tactic to prove the explicit sign.

Layer 5. Moment extension. Separately formalize the beta-integral identity in [lemma 6.1](#), then derive integral positive definiteness of K_d .

A schematic Lean data structure is:

```
structure ExpNode where
  m : Nat
  n : Nat

def inputNode (A B : Nat) (e : ExpNode) := A^e.m * B^e.n
def outputNode (e : ExpNode) := 2^e.m * 3^e.n

def certificateNumerator (A B : Nat) (E : Fin (k+1) -> ExpNode) : Int :=
  Matrix.det (fun i j =>
    if j < k then (inputNode A B (E i) : Int)^j
    else (outputNode (E i) : Int))
```

The final contradiction theorem takes as input the sortedness proof and a normalized integer value for the determinant. This makes discovered certificates auditable even before the full search is formalized.

14 Progress assessment and next attacks

The new framework changes the main question from “How can one estimate another scalar divided difference?” to two more structured problems.

Open problem 14.1 (Uniform mixed-sign problem). Show that for every integer pair (A, B) satisfying the elementary order constraints but not of the form $(2^r, 3^r)$, some primitive finite exponent pattern has a determinant sign contrary to (5); preferably with an order bounded effectively in terms of $\log A + \log B$.

Open problem 14.2 (Moment-lattice gap problem). Construct mixed node sets for which either a principal minor of K_d is nonpositive, or an analytic estimate proves $\det K_d < 1$. Equivalently, find an integer polynomial violating (22) by a provable Stieltjes concentration estimate.

The immediate research program is:

- (1) enumerate primitive exponent patterns modulo the scaling law (6);
- (2) factor low-order mixed determinant numerators symbolically and identify divisibility that distinguishes common-power controls;

- (3) compute Smith invariants and subresultant remainder signs for initial smooth-node sets;
- (4) optimize integer-Chebyshev polynomials against the exact normalized measure, with rigorous tail bounds from [lemma 9.3](#);
- (5) convert any recurring negative minor into a small formal certificate and search for a parameter-uniform factorization.

The strongest proved advance is [theorem 7.1](#) and [corollary 7.2](#): a hypothetical counterexample must support, for every admissible mixed node set, a positive definite *integral* Hankel form whose normalized coefficient lattice has a quantified minimum. The strongest negative advance is [theorem 4.1](#): the consecutive pure-axis sign hierarchy is automatic and cannot settle compatibility. Together these facts focus subsequent effort on output-sensitive mixed minors and their analytic concentration.

A A compact dependency graph of the main results

Result	Dependencies
theorem 3.1	reciprocal normal form; generalized mean-value theorem; generalized Vandermonde identity
proposition 3.2	homogeneity of u^θ and Vandermonde scaling
theorem 4.1	standard Vandermonde determinant in the column bases $1, A, \dots, A^{k-1}, 2$
lemma 6.1	beta integral; partial fractions; analytic continuation within the convergence strip
theorem 7.1	lemma 6.1 ; divisibility of $\Delta/P'(u_i)$; positivity of a square integral
corollary 7.2	integral positive definiteness and normalization by k_0
proposition 8.1	Laurent expansion of R/P ; classical Padé–Hankel–subresultant identities
lemma 9.3	pointwise comparison of the Stieltjes weight under dilation

B Sanity checks against integral exponents

For $x = r \in \mathbb{N}$, one has $A = 2^r$, $B = 3^r$, and $\theta = 1/r$. Every theorem above must remain consistent with these controls. In particular:

- the determinant signs in [theorem 3.1](#) hold strictly;
- the axis factorizations are exact;
- every K_d is integral positive definite;
- no valid analytic estimate can make $\det K_d < 1$;
- any proposed “universal” localization bound that also excludes these controls has lost essential mixed arithmetic information.

This control family is built into the exact search and is a practical defense against an incorrect sign, a missing Vandermonde factor, or a non-rigorous numerical Cholesky step.

C Bibliographic orientation

The conjecture originates in the work of Alaoglu and Erdős on highly composite and related numbers [1]. Its transcendence-theoretic setting is the Four Exponentials Conjecture and the proved Six Exponentials Theorem; standard references include [10, 12, 3]. The Gelfond–Schneider reduction used in [proposition 2.1](#) is classical [7, 3]. Bernstein and Stieltjes representations are treated systematically in [11, 2]; total positivity and sign-regular kernels in [8]. The q -Newton notation in (9) follows [6]. Padé, continued-fraction, determinant, and subresultant techniques are covered in [13, 4, 9]. Exact real-algebraic and fraction-free certificate methodology is consonant with [5].

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